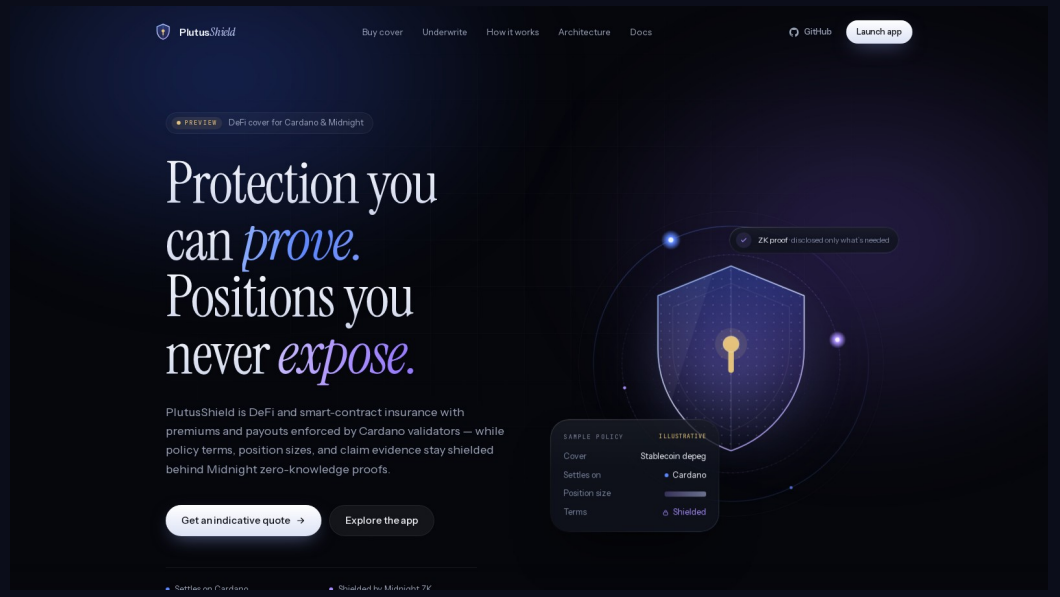
MIDNIGHT BUILDATHON · WAVE 2

# Protection you can prove. Positions you never expose.




PlutusShield is DeFi and smart-contract insurance. Cardano holds the money and pays claims. Midnight keeps who is covered, for how much, and the claim evidence private.

kshot3000.github.io/PlutusShield · github.com/Kshot3000/PlutusShield

THE PROBLEM

# DeFi cover today makes you choose: get paid, or stay private.

## Cover leaks your book

On a public chain a policy says who you are, what you hold and how much you insured. That is a map for anyone hunting large positions.

## Claims leak your evidence

Exploit claims need proof: tx traces, screenshots, wallet history. Posting it publicly exposes the claimant and sometimes the bug.

## Payouts need trust

Most cover relies on one assessor or a token vote. Users need to see that no single key can approve or block a payout.

HOW IT WORKS

# Cardano settles. Midnight keeps the secrets.

## Cardano Preview · Aiken, Plutus V3

Premiums paid in ADA or test USDC (tUSDCx) into on-chain pools.

Depeg cover pays from oracle feeds; sales close by themselves if the feeds go stale or the peg breaks.

Exploit cover pays only with 2-of-3 committee signatures, enforced by the validator.

The policy datum holds a hash commitment, not the terms.

## Midnight Preprod · Compact, ZK

registerPolicy stores a commitment to the holder and terms.

proveCover(policyId, minCoverage) proves cover is at least X without revealing the real limit.

fileClaim puts only a commitment to the encrypted evidence on chain.

voteClaim tallies three committee seats; 2 approvals mark the claim PAID.

ARCHITECTURE

# One policy, two chains, a relay in between.




No wallet address, position size or evidence lands on Midnight in the clear. The relay only moves commitments one way and the committee decision the other.

BUILT ON MIDNIGHT

# Two Compact contracts, deployed and exercised on Preprod.

## policy-cover v1

Preprod contract `d84c6187…925e8787`

Circuits: registerPolicy, proveCover, rotateHolder, fileClaim, resolveClaim, expirePolicy.

5 policies registered and proven, 4 claims filed and resolved.

Single assessor: the first version, kept for the record.

## policy-cover v2

Preprod contract `5430b803…4feb8110`

Adds voteClaim: a 2-of-3 committee with per-round tallies; a reject resets the round.

First drill: approve, reject, approve, then PAID at 2 of 3.

Second: a holder-filed claim (claim kit), seats 1 + 3 approve, paid on Cardano.

4 Cardano Preview policies registered on v2.

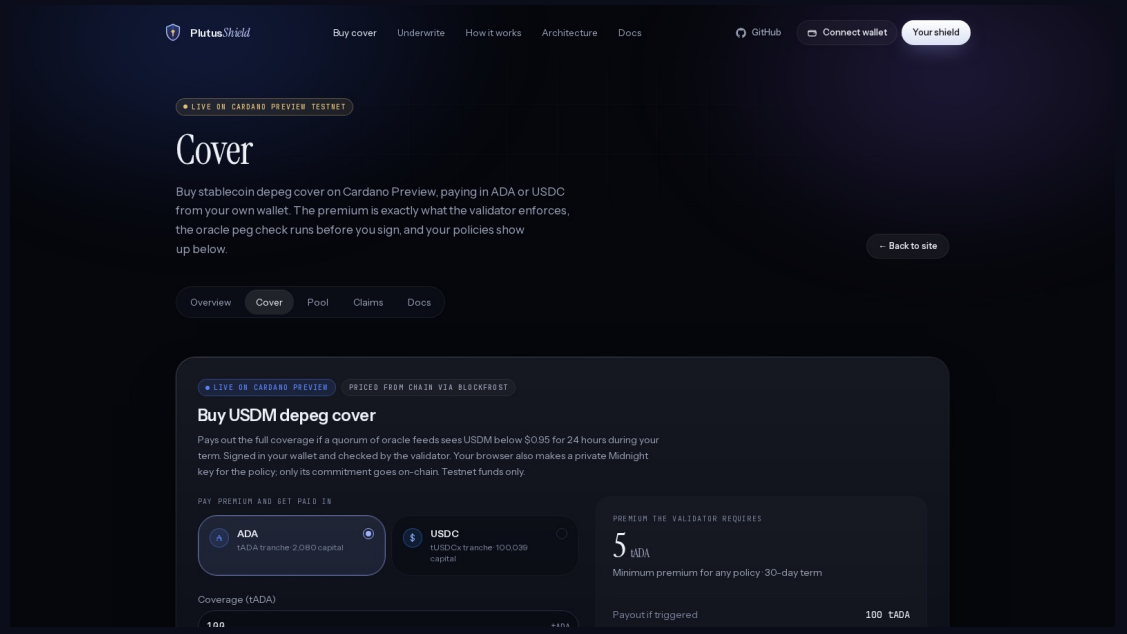
`compactc 0.31.1 · both contracts full-compile with prover and verifier keys`

LIVE ON TESTNET

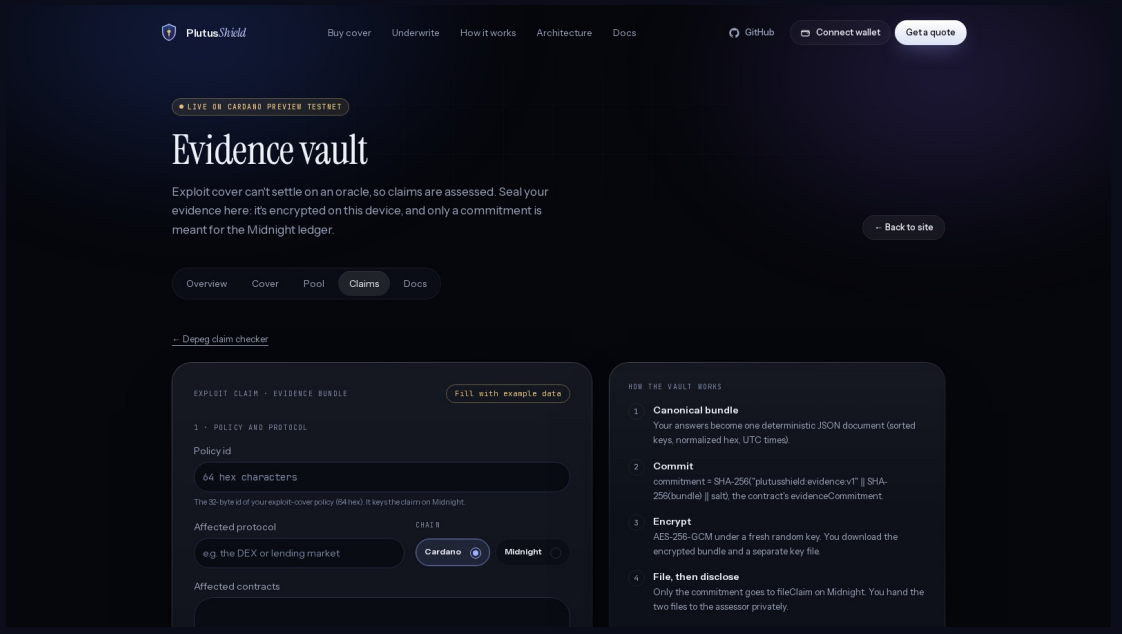
# Every line here is backed by a transaction.

| | |
|---|---|
| Depeg pool (ADA + tUSDCx) | 11 live Preview policies, buy / claim / release from a CIP-30 wallet |
| Exploit cover in the browser | 10 tADA bought with the website's own Buy builder, Buy 3c1fc629… |
| First exploit claim paid | 50 tADA: Midnight claim PAID, then Cardano Settle 5509c2d9… |
| First committee payout | 50 tADA co-signed by 2 of 3 assessors, Settle c35535dd… |
| Holder-filed claim paid | Claim kit fileClaim 00989a6a…, v2 voteClaim 2 of 3, 25 tADA Settle dbf0ba7a… |
| 1-of-3 payout attempt | Rejected by the validator in both local and Blockfrost evaluation |
| Depeg oracle | 52 recorded live publishes to 3 Preview feeds, latest 99.97% (Oct 12) |
| Midnight relay | registerPolicy + proveCover on Preprod (v1 + v2), readable via Blockfrost |

# A real app: quote, buy, prove, claim.




/cover: connect a Preview wallet, price depeg or exploit cover, pay in ADA or tUSDCx.




/claim/evidence: evidence is encrypted in the browser (AES-256-GCM); only its SHA-256 commitment goes to fileClaim.

ENGINEERING

# 344 unit tests plus an end-to-end emulator run, all green.

**131**

Aiken validator tests

**136**

SDK: quote engine, risk tiers, codecs, evidence vault, claim kit, Lace, prove bridge

**60**

Compact circuit simulation, claims relay, relay plan, state decode

**14 + 3**

oracle relay · committee kit

Lucid emulator e2e: pool, buy, settle, expire, exploit 2-of-3, every 1-of-3 combination rejected, exploit release.

Next.js static site, 18 routes, redeployed to GitHub Pages on every push; checked on iPhone (WebKit) and Pixel.

WHERE IT STANDS

# Honest about today. Clear about next.

## Today

Testnet only (Cardano Preview, Midnight Preprod). Not audited.

The three committee seats are run by the team.

Exploit evidence in the drills is demo evidence.

Holders file claims with the claim kit; File with Lace proves locally, first finalized submit pending.

## Next

Independent assessor operators (offline keygen and signed enrollment kit is in the repo).

First finalized File with Lace submit: the holder prove bridge already proves proveCover / fileClaim in about 1 s; Lace balances and submits.

Security audit, then a capped mainnet launch.

Catalyst funding for claim capital.

TRY IT

# Insurance that pays on Cardano and stays quiet on Midnight.

Live app https://kshot3000.github.io/PlutusShield/

Code https://github.com/Kshot3000/PlutusShield

Built by @kshot9000